

Workup for 3d grating dispersion diagrams (s and p polarizations)

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- Working to recreate figure 1c in [1]
- Plots made by doing an array job with each array entry being a meep [2] simulation with one frequency and angle of incidence combination.
- Array was generated from the outer product of two linspaces:

```
def expand_grid(data_dict):  
    rows = itertools.product(*data_dict.values())  
    return pd.DataFrame.from_records(rows, columns=data_dict.keys())  
  
args = expand_grid(  
    {  
        "angle_deg": np.linspace(0, 20, 30),  
        "freq_THz": np.linspace(220, 290, 50)  
    }  
)
```

- Each job in the array runs the same transmittance function:
 1. Set cell dimensions to $0.6 \times 0.6 \times (2 \times \text{dpml} + 2 \times \text{dpad} + \text{ph})$
 2. Set a source at the top of the cell, spanning the whole area
 3. Set the source to have a phase that is spatially dependent, call `k_point` to take the frequency, index, and angle into account so that the periodic boundaries enforce the phase changes

```
def pw_amp(k, x0):  
    def _pw_amp(x):  
        return cmath.exp(1j * 2 * math.pi * k.dot(x + x0))  
    return _pw_amp
```

```
k_point = mp.Vector3(0, 0, fcen * 1.45).rotate(mp.Vector3(0, 1, 0), theta_r
```

4. Set a monitor at the bottom of the cell, spanning the whole area
5. Run a simulation of just silica (index of 1.45)

6. Store the monitor flux
 7. Create the silicon post geometry with radius pr, height ph, axis $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, index of 3.67
 8. Run a simulation with this post included
 9. Store the monitor flux
 10. Divide the second monitor flux by the first to get transmittance
- s polarization computed with Ey, p polarization computed with Hy

References

- [1] Y. Zhou, H. Zheng, I. I. Kravchenko, and J. Valentine, “Flat optics for image differentiation,” en, *Nature Photonics*, vol. 14, no. 5, pp. 316–323, May 2020, issn: 1749-4893. DOI: 10.1038/s41566-020-0591-3. Accessed: Feb. 26, 2026. [Online]. Available: <https://www.nature.com/articles/s41566-020-0591-3>.
- [2] A. F. Oskooi, D. Roundy, M. Ibanescu, P. Bermel, J. D. Joannopoulos, and S. G. Johnson, “Meep: A flexible free-software package for electromagnetic simulations by the FDTD method,” *Computer Physics Communications*, vol. 181, no. 3, pp. 687–702, Mar. 2010, issn: 0010-4655. DOI: 10.1016/j.cpc.2009.11.008. Accessed: Feb. 27, 2026. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S001046550900383X>.